

# Session Objectives

- 1. Brief review of normal muscle and, inflammatory and non-inflammatory muscle reactions and disorders such as degeneration, atrophy, etc.
- 2. Describe, discuss, and compare the pathology/pathogenesis including etiology, morphology, and clinicopathological manifestations of **Spinal Muscle Atrophy (Werdnig-Hoffman) disease, Muscular Dystrophies, and Myotonic dystrophy**
- 3. Describe and discuss the pathology/pathogenesis including etiology, morphology, and clinicopathological manifestations of **malignant hyperthermia (ion-channel myopathy), congenital myopathies, disorders of lipid/glycogen metabolism and mitochondrial myopathies.**
- 4. Describe, discuss and compare the pathology/pathogenesis including etiology, morphology, and clinicopathological manifestations of the major inflammatory myopathies which include **dermatomyositis, polymyositis, and inclusion body myositis.**
- 5. Describe and discuss the pathology/pathogenesis including etiology, morphology and clinicopathological manifestations of **myasthenia gravis and Lambert-Eaton myasthenic syndrome**

# Skeletal Muscle Pathology

- Brief review of normal skeletal muscle
- Muscle reactions/injury
- Inherited Diseases of Skeletal Muscle
  - Spinal muscular atrophy (Werdnig-Hoffman disease)
  - Muscular Dystrophy (Duchenne and Becker)
  - Myotonic dystrophy
  - Malignant hyperthermia (Ion channel myopathy)
  - Congenital myopathies
  - Diseases of Glycogen/Lipid Metabolism
  - Mitochondrial myopathies
- Inflammatory myopathies (polymyositis, dermatomyositis, inclusion body myositis)
- Myasthenia Gravis and Lambert-Eaton Syndrome
- Miscellaneous myopathies

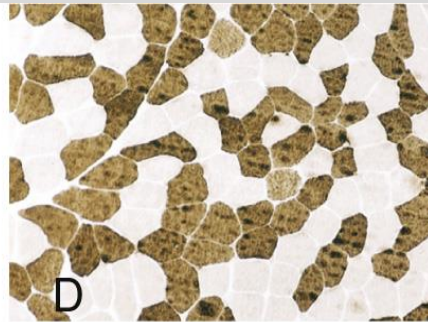
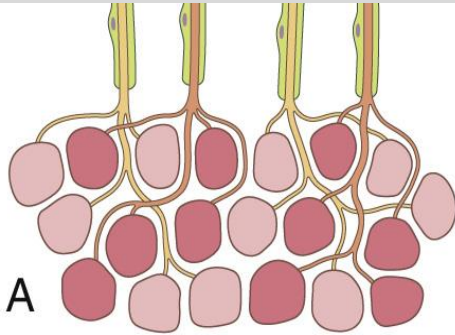
# Myogenic Injury

- Segmental Myofiber Degeneration and Regeneration
  - Occurs when **part of a myofiber undergoes necrosis**
  - Degeneration
    - **Creatine kinase released into blood from damaged cells**
    - Myophagocytosis occurs (macrophages remove damaged components of the myofiber)
  - Regeneration
    - Begins with fusion of an **activated satellite cell** to the damaged myofiber
    - **Regenerated myofibers stain basophilic** due to high levels of RNA
    - In chronic disease, complete regeneration is not possible due to the **disease process outpacing regeneration**
- Myofiber hypertrophy is typically a **physiologic response to exercise** but can be associated with a disease processes
- Cytoplasmic inclusions include vacuoles, protein aggregates, clustered organelles

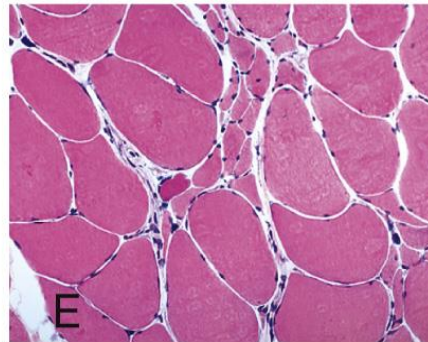
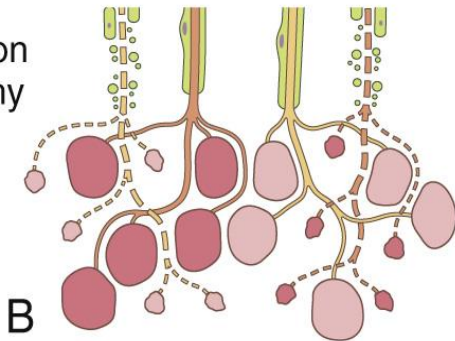
Robbins and Cotran Pathologic Basis of Disease, 10<sup>th</sup> ed. 2021, Ch. 27

# Neurogenic Injuries

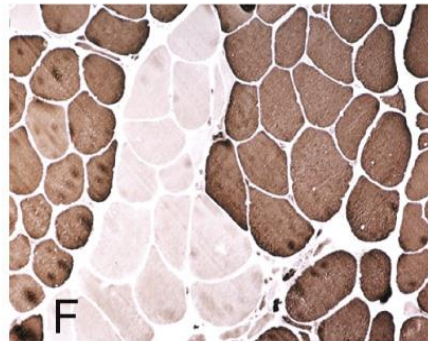
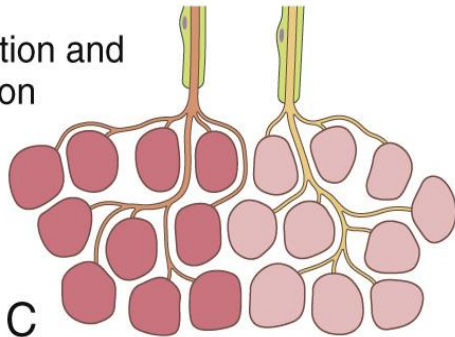
Normal



Denervation and atrophy



Reinnervation and regeneration



- **Disruption of innervation** of a muscle that causes muscle dysfunction (usually occurs as **grouped atrophy, E**)
- Mechanism: **innervating motor neuron determines fiber type**
  - Denervation causes **atrophy** of those fibers as a **group** (**flattened, angular shape**)
  - Reinnervation will normalize the shape but may **change the fiber type**; axons belonging to unaffected nearby motor unit extend sprouts to reinnervate myocytes; newly reinnervated fibers become fiber type of that motor unit: loss of checkerboard pattern, D (histologically motor units appear as groups of myofibers of the same histochemical type (fiber type grouping, F))

# Spinal Muscular Atrophy: SMA I

- Disorders with childhood onset, progressive weakness
  - **SMA I = Werdnig-Hoffman's Disease**; most severe
- Pathogenesis: **Autosomal Recessive**: loss of fxn mutation in SMN1 gene (chrom 5) [possible role in RNA splicing (snRNP assembly)] – causes defects in spliceosomes
  - **Loss of motor neuron innervation (anterior horn cell destruction/degeneration) → muscle weakness and atrophy**
  - **Degeneration of the anterior horns (motor neurons) of the spinal cord**
  - **Denervation of skeletal muscle**
- Clinical features –LMN signs
  - Infants: can present with **generalized hypotonia (“floppy infant”)-little muscle tone, presents within 4 months of birth; death usually by 2 - 3 years**
- Diagnostic features - biopsy
  - Histology: large areas of flattened, angulated, very atrophic myofibers mixed with normal/hypertrophied myofibers in small groups (those that retain innervation)

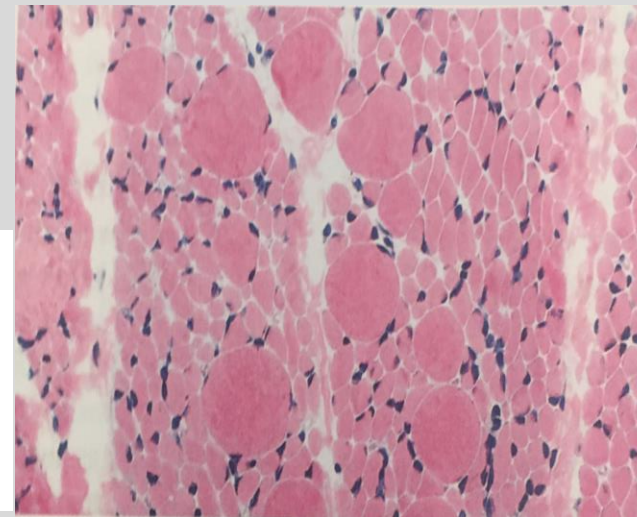


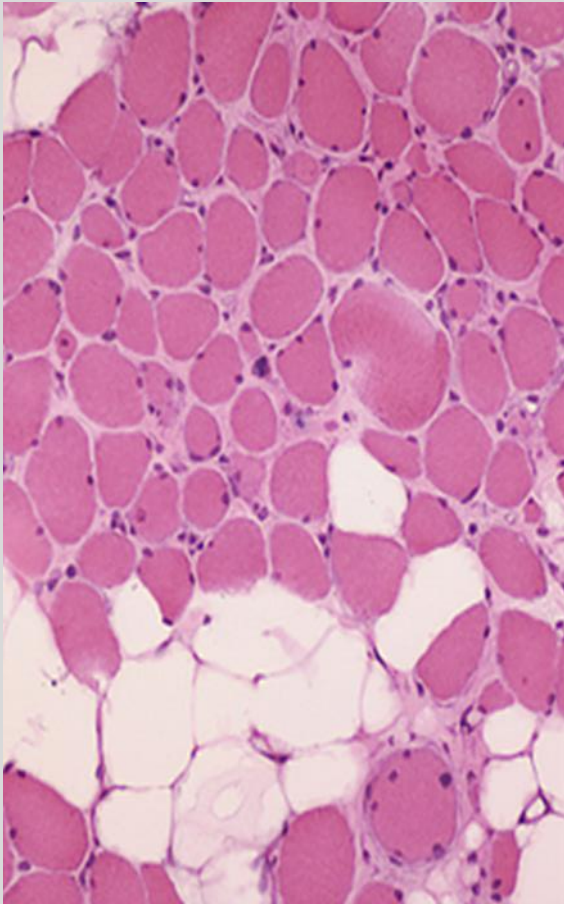
Figure 27. 13 Robbins and Cotran Pathologic Basis of Disease, 10<sup>th</sup> ed. 2021, Ch. 27  
First Aid 13<sup>th</sup> ed. 2020



# DMD Histopathology

- **Early disease:**

- muscle biopsies in young boys: ongoing muscle damage, i.e., **segmental myofiber degeneration and regeneration** associated with admixture of **atrophic myofibers**
- **myophagocytosis, usually no inflammation**
- **fascicular architecture still preserved**



- **As DMD progresses:**

- muscle tissue **replaced by collagen and fat cells** (“fatty infiltration”, image) - causes pseudohypertrophy of calves
- remaining myofibers: **prominent variation in size** (small atrophic fibers to large hypertrophied fibers)
- **fascicular architecture distorted, abnormal**

# DMD

- Clinical features: Usually **Proximal muscle weakness and clumsiness**
  - **Pseudohypertrophy of the calf muscles:** enlargement of the muscle with associated weakness (**muscles are replaced by collagen and fat tissue**)
  - **Gower's Sign:** use of their hands to stand from a lying/sitting position
  - Later, develop joint contractures, scoliosis, worsening respiratory reserve and sleep hypoventilation
  - Can develop cardiomyopathy and arrhythmias (protein is located primarily in skeletal muscles and in heart, small amounts of dystrophin are present in nerve cells in brain)
  - May have intellectual impairment
  - Usually increased serum creatine kinase (early stages)
  - **Death may result from respiratory insufficiency by the 20s – 30s**
- Treatment: supportive; New efforts to manipulate RNA splicing, promotion of ribosomal “read-through” of stop codons

# Ion Channel Myopathies

- These **skeletal muscle channelopathies** are inherited mutations affecting function of ion channel proteins
  - Some increase excitability (hypertonia), some decreased excitability (hypotonia)
  - Some are subclassified based on K levels (examples, hyperkalemic, or hypokalemic, or normokalemic periodic paralysis)
  - Depending on affected ion channel, clinical manifestations may include:
    - epilepsy, migraines
    - movement disorders with cerebellar dysfunction
    - peripheral nerve disease, muscle disease



# Ion Channel Myopathies cont.

- Some examples of mutations assoc. with muscle dysfunction:
- CACNA1S protein is subunit of muscle calcium channel and a missense mutation most common cause of **hypokalemic paralysis**
- CLC1 mutation affects chloride channel, associated with myotonia congenita, and **CLC1 expression decreased in myotonic dystrophy**
- **Mutations RYR1 → disrupt function of ryanodine receptor (regulates calcium release from the sarcoplasmic reticulum) and are involved in congenital myopathy (central core disease) and malignant hyperthermia**

# Malignant Hyperthermia

- Rare but life-threatening **skeletal muscle channelopathies**  
Pathogenesis: **Autosomal dominant: Mutation in the voltage-gated RYR1 gene affects ryanodine-r (involved with controlling release of Ca<sup>2+</sup> from sarcoplasmic reticulum)**
  - **Increased calcium release** from sarcoplasmic reticulum following administration of **halogenated inhaled anesthetics and succinylcholine (temporary paralysis, rapid onset and short duration of muscle relaxation)**
  - **Mechanism: Interaction of anesthetic and mutated receptor causes incr. calcium efflux which leads to tetany and heat production**

# Malignant Hyperthermia clinical features

- **Hypermetabolic state**
  - **Hyperthermia** and later, hyperpyrexia (temp > 106 degrees F)
  - Severe muscle contractions (**tetany**)
  - **Tachycardia/Tachypnea**
- May occur in predisposed patients with **hereditary muscle disease**, inc. congenital myopathies, dystrophies, and metabolic myopathies
- Treatment: **Dantrolene** (ryanodine receptor antagonist)

# Disorders of Lipid or Glycogen Metabolism

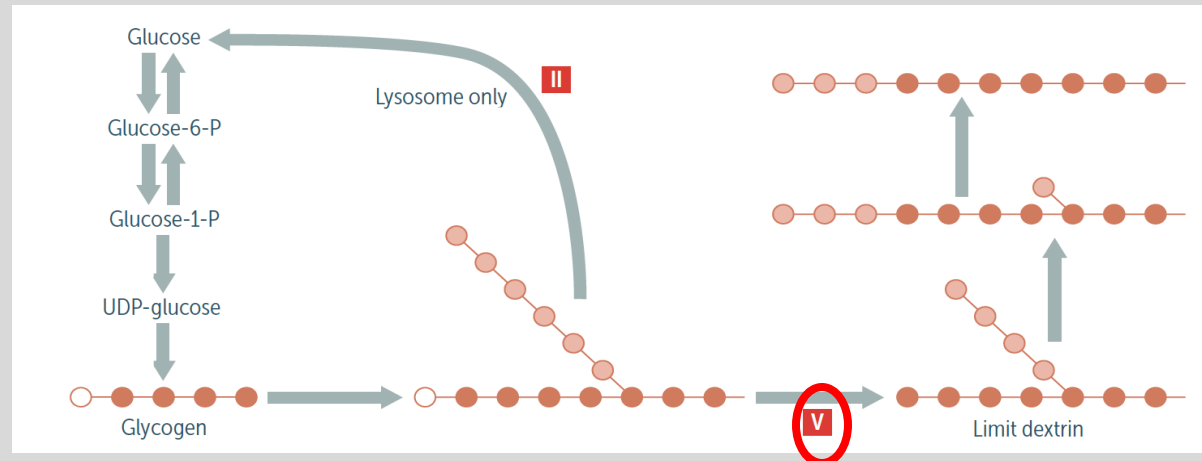
- General patterns of muscle dysfunction:
  1. Symptomatic with exercise/fasting which range from muscle cramping to rhabdomyolysis
  2. OR slowly progressive muscle damage, no episodic symptoms
- Examples (more details follow)
- **Carnitine palmitoyltransferase II deficiency**
  - Defect in free fatty acid transport into mitochondria
- **Myophosphorylase deficiency (McArdle disease)**
  - Episodic damage, eg. myoglobinuria
- **Acid maltase deficiency (Pompe disease)**
  - Impaired lysosomal conversion glycogen to glucose

# Carnitine Palmitoyltransferase II Deficiency

- **Most common disorder of lipid metabolism**
- Pathogenesis: Enzyme deficiency
  - Impaired transport of free fatty acids into the mitochondria
  - **Toxic build up of fatty acids in the cytosol**
- Clinical features: Episodic muscle damage with exercise or fasting
  - **Hypotonia**
  - Hypoketotic hypoglycemia
  - Dilated cardiomyopathy

# Diseases of Glycogen Metabolism

## McArdle Disease

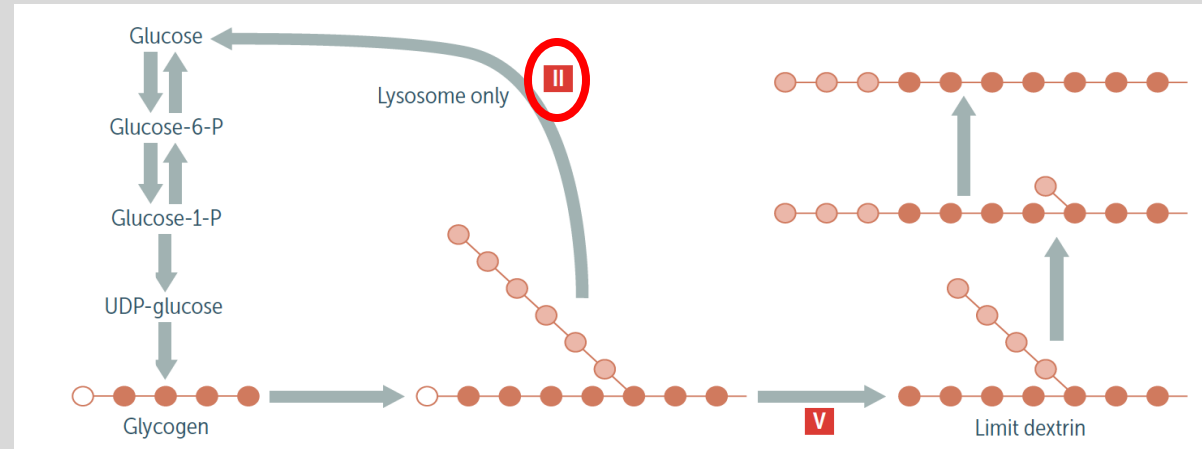


- Pathogenesis: **Myophosphorylase (skeletal muscle glycogen phosphorylase) deficiency (type V)**
  - **Inability to break down glycogen in muscle**
- Clinical features: Episodic muscle damage with exercise
  - **Painful muscle cramps**
  - **Myoglobinuria**
  - **Arrhythmia** (due to electrolyte imbalances)
  - **Normal blood glucose levels** (liver is unaffected)
- Diagnosis: Venous lactate-ammonia test, during exercise, **venous lactate curve is flat** with a normal rise in ammonia levels
- Treatment: Limit carbohydrates rich diet and careful exercise routine



# Diseases of Glycogen Metabolism

## Pompe Disease



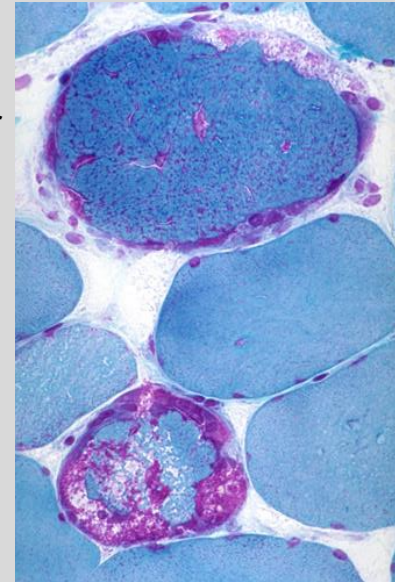
- Pathogenesis: **Acid maltase (lysosomal acid  $\alpha$ -1,4-glucosidase) deficiency (type II)**
  - Impaired glycogen breakdown within the lysosome
- Clinical features: Heart, liver and muscle can be affected
  - **Cardiomegaly** (Hypertrophic cardiomyopathy)
  - **Hypotonia** (severe deficiency presents in infancy)
  - Exercise intolerance
  - Mild deficiency affects respiratory and truncal muscles
  - Early death is seen in patients with systemic symptoms
- Treatment: Enzyme replacement therapy works for some patients

# Mitochondrial Myopathies

- Rare disorders; may involve many organ systems
- Pathogenesis: Involves mutations of **mitochondrial DNA** with **maternal inheritance** because only oocyte contributes mitochondria to embryo. Disease is thought to be due to **mutations that impair ATP generation** in mitochondria
  - **All children of an affected female** may show signs of disease due to maternal transmission (affected males can't transmit to their children), has varying degrees of expressivity
  - Tend to **affect skeletal muscles/other tissues with high ATP requirements (cardiac muscle cells-dysrhythmia, neurons-seizures)**, drug therapy for MERRF syndrome includes metabolic drugs involved with or improve cellular respiration (ex. metformin, creatine, and L-carnitine), clonazepam for myoclonus and anticonvulsants
  - Skeletal muscle involvement causes weakness, increase in serum **CK**, or **rhabdomyolysis**
  - Weakness in certain regions is variable, but **involvement of extraocular eye muscles common**
  - Chronic progressive **external ophthalmoplegia** (inability to move eyes, eyelids and eyebrows – blurred, double vision) is common feature of mitochondrial disorders

# Mitochondrial Myopathies

- **Myoclonic epilepsy with ragged red fibers (MERRF)**
  - **Multisystem** mitochondrial diseases **characterized by predominantly intermittent muscle twitches aggravated by action (progressive myoclonus**, hiccups are a form of myoclonus, also the sudden jerks, or "sleep starts," you may feel just before falling asleep - these forms of myoclonus occur in healthy people and rarely present a problem), **weakness (myopathy – limb girdle type), and progressive stiffness (spasticity)**, diagnosis/biopsy: "ragged red fibers" due to accumulation of affected mitochondria in the subsarcolemmal area of myofibers, increased **serum creatine kinase**
  - Most also experience other types of seizures in addition to myoclonus, may be of generalized tonic-clonic, atonic or absence types
  - Modified **Gomori trichrome** stain allows visualization of the mitochondrial activity and collection by staining red (ragged red fibers), Pathogenic variant in mitochondrial gene MT-TK is present in more than 80% of affected
- **Leber hereditary optic neuropathy**
  - **Predominantly** degeneration of retinal ganglion cells and their axons leads to an acute or subacute loss of central vision; it primarily affects young adult **males**; different mitochondrial mutation from MERRF
  - Commonly present with chronic progressive external **ophthalmoplegia** (weakness of the extraocular eye muscles)



Mitochondrial clusters appear as red deposits in cryostat sections of muscle stained with modified Gomori trichrome  
<https://neuropathology-web.org/chapter10/images10/10-rrfl.jpg>

# Dermatomyositis

- Rare inflammation of skin and skeletal muscle; **Immune-mediated** small blood **vessel** injury leads to **muscle damage**, peak age onset **50-60 years**, more common in **females**, gradually worsens over weeks to months, can last several years or be lifelong
  - Classic skin rash: **Lilac** or **heliotrope** (reddish-purple rash on or around the eyelids) with **periorbital edema**
  - May have **Gottron's lesions/papules** (scaling erythematous eruption or dusky red patches on knuckles, elbows, or knees).
    - **Proximal muscles weak first; symmetric**, slow onset, **myalgias**
    - Dysphagia may develop if oropharyngeal, esophageal muscles involved
  - **Extramuscular involvement** includes interstitial lung disease (10%) vasculitis and myocarditis
  - 15 - 24% of adults with dermatomyositis may have **malignancy** (Lung, breast, ovary, colorectal, cervical, bladder, nasopharyngeal, esophageal, kidney, lymphatic, hematopoietic)

# Dermatomyositis

- Pathogenesis: **Autoimmune** disease characterized by small vessel damage, **capillaries** attacked by Abs and complement causing foci of **ischemic myocyte necrosis**
- Although the cause of dermatomyositis is unknown, several genetic, immunologic, and environmental factors are implicated, having another autoimmune disease is a risk factor
- **Anti-Mi2 Abs** (anti-helicase) associated with **Gottron papules** and **heliotropic rash**
- **Anti-Jo1 Abs** (anti-histidyl t-RNA synthetase): associated with **interstitial lung disease, non-erosive arthritis** and **“mechanic’s hand” rash**

# Polymyositis

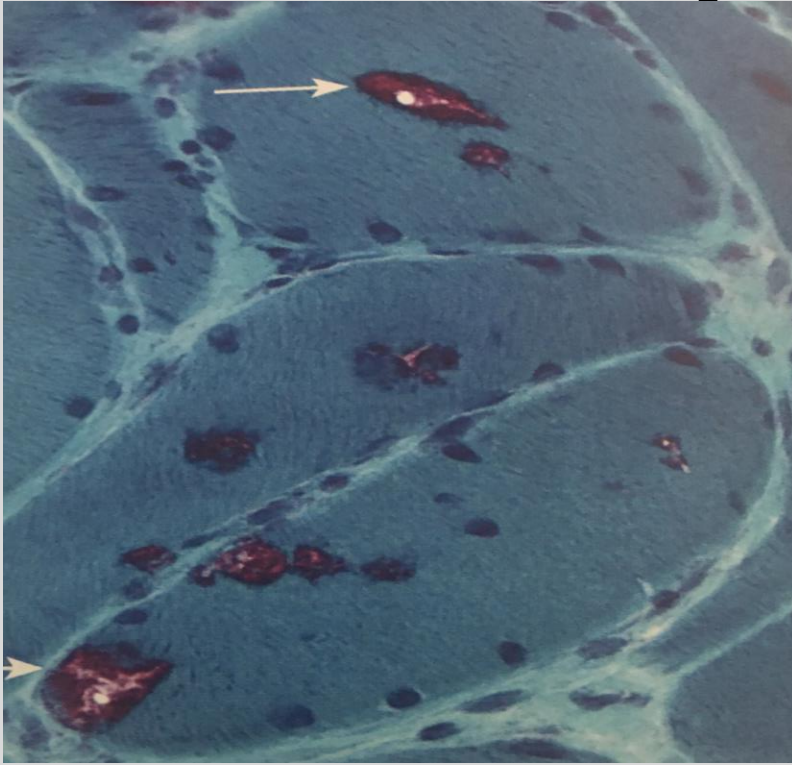
- Symmetric proximal **muscle** involvement **similar to dermatomyositis but no skin involvement**, occurs mainly in **adults**, (peak age onset 30-60 years) and may have inflammatory involvement of **heart, lungs, and blood vessels**, women are affected more often than men, can last several years or be lifelong
- Pathogenesis: Immune-mediated (**CD8+ cytotoxic T cells**) **tissue damage, there is no vascular injury** (in contrast with dermatomyositis)
- Polymyositis is not caused by a genetic mutation, but some genes may influence the risk of developing polymyositis, having another autoimmune disease is a risk factor, exact cause unknown, but has characteristics of autoimmune disorders, which occur when the immune system mistakenly attacks healthy body tissues
- Histology: **Endomysial inflammation** (predominately CD8+ T cells)
  - Necrotic and regenerating muscle fibers scattered throughout fascicle
  - **No perifascicular atrophy** (seen in dermatomyositis); no evidence of vascular damage
- Clinical Findings: **myalgia** + symmetric **proximal muscle** weakness
- Labs: increased serum **creatine kinase**
- Treatment: immunosuppressive agents

# Inclusion-Body Myositis

- **Most common inflammatory myopathy of patients age > 65yo**
- Pathogenesis: Unclear if inflammatory or degenerative process
- Clinical Findings:
  - Begins with **distal muscle involvement**, may be **asymmetric** (v. dermatomyositis and polymyositis)
  - Slowly progressive muscle weakness most severe in **quadriceps and distal upper extremities**, may have dysphagia



# Inclusion Body Myositis



Gomori trichrome stain

## Diagnosis/Labs:

- Moderately increased creatine kinase levels
- Can be **cytosolic 5'-nucleotidase 1A (cN1A) antibody positive (50% of patients)**
- Histology: **“rimmed vacuoles”** (autophagic vacuoles containing degrading organelles and other cytoplasmic contents) **in myofibers**
- CD8+ T cell endomysial infiltration similar to polymyositis

Treatment: Immunosuppressive agents or IVIG (**not as effective as in other inflammatory myopathies**)

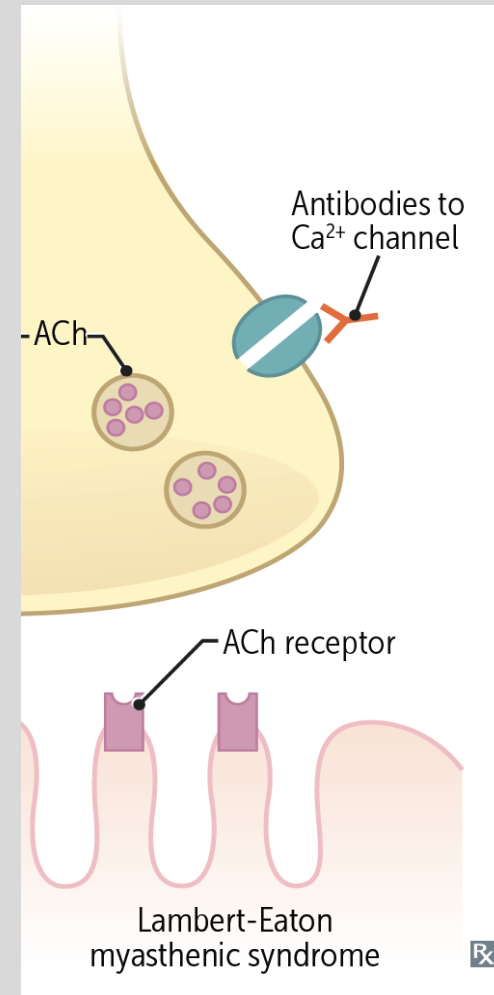
Robbins and Cotran, Pathologic Basis of Disease, 10<sup>th</sup> ed 2020

# Myasthenia Gravis

- Clinical features:
  - Bimodal distribution; young **females (20-40 years)**, older **males (50-70 years)**, more commonly affects women (2:1), if arising before age 40 years
  - **Painless weakness, fatigue—worsens with repeated exertion**
  - Weakness in proximal muscles, diaphragm, neck extension
  - **Diplopia, ptosis**
  - Dysphagia for both solids and liquids
  - Reflexes spared
  - **Thymic abnormalities associated** (thymoma, thymic hyperplasia)—esp in younger pts (10% of patients have a thymoma, 30% of patients (usually younger) have thymic hyperplasia)
  - **Electrophysiologic studies** show decreased muscle response with repeated stimulation
  - **Nerve conduction** studies normal
- Treatment: **Acetylcholinesterase inhibitors** are first line (inhibit acetylcholinesterase which increases acetylcholine and reverses muscle weakness)
  - Others: plasmapheresis, immunosuppressive drugs, thymectomy (if thymoma)

# Lambert Eaton Myasthenic Syndrome

- **Rare** autoimmune disease, more common in **males**, onset age peaks at **35 years**, with a larger peak at **60 years**
- Pathogenesis: autoantibodies against **pre-synaptic calcium channels that prevents the release of ACh**
- Clinical features: Muscle weakness that **improves with exertion**
  - **Autonomic symptoms** (dry mouth, constipation, impotence)
  - Hyporeflexia
  - 50% of patients have an **associated malignancy** (**neuroendocrine lung carcinoma, e.g., small cell carcinoma**)
  - Patients may have other autoimmune diseases as well
- Diagnosis: electrophysiologic studies
  - Repetitive stimulation increases the muscle response (**opposite of MG**)
- Treatment:
  - Drugs that increase the release of ACh into the synapse
  - Immunosuppressive agents



# Summary Slide 1

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- **SMA1 (Werdnig-Hoffman disease)** - autosomal recessive; degeneration of anterior horns (motor neurons) of spinal cord; loss-of-function mutation in SMN1 gene causes defects, “floppy” baby (hypotonia)
- **Duchenne muscular dystrophy** - X-linked recessive, frameshift deletion or nonsense mutation of dystrophin gene, **absent dystrophin protein**; early childhood, increased CK; proximal muscle weakness and clumsiness; **pseudohypertrophy of calf muscles**: (“fibrofatty infiltration”), **Gower’s sign**, cardiomyopathy, arrhythmias, intellectual impairment
- **Becker muscular dystrophy** - X-linked recessive, loss-of-function mutation of dystrophin gene results in truncated, **partially functional dystrophin protein, less severe**, slower progression than DMD
- **Myotonic dystrophy** - autosomal dominant, **CTG trinucleotide repeat** in DMPK gene, **CLC1 deficiency**, myotonia, muscle wasting, cataracts, frontal balding, testicular atrophy, etc.
- **Malignant hyperthermia** - autosomal dominant with variable penetrance; mutation ryanodine receptor (RYR1) gene results in increased calcium release from sarcoplasmic reticulum triggered by halogenated inhaled anesthetics and succinylcholine: hypermetabolic state (hyperthermia, muscle spasms, tachycardia, etc.)
- **Congenital myopathies- Central Core Disease** (AD, mutation of RYR1 gene, early onset hypotonia, skeletal abnormalities, cytoplasmic cores)/ **Nemaline Myopathy** (AD or AR, NEM 1-7 gene mutations; childhood weakness/“floppy” baby, Nemaline rods on biopsy)

# Summary Slide 2

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- **Diseases of lipid metabolism - Carnitine Palmitoyltransferase II Deficiency;** impaired transport of **free fatty acids** into mitochondria: build up of fatty acids in cytosol, episodic muscle damage with exercise or fasting, hypotonia, hypoketotic hypoglycemia, dilated cardiomyopathy
- **Diseases of glycogen metabolism:**
  - **McArdle disease** (Myophosphorylase deficiency (type V), **episodic muscle damage with exercise, painful muscle cramps, myoglobinuria, normal glucose**)
  - **Pompe disease** (acid maltase deficiency type II; impaired glycogen breakdown within lysosome; cardiomegaly/hypertrophic cardiomyopathy, hypotonia, exercise intolerance)
- **Mitochondrial myopathies** -mitochondrial inheritance (maternal transmission); mutations **impair ATP generation** in mitochondria; myopathy, lactic acidosis & CNS disease; MELAS syndrome; fatigue, weakness, rhabdomyolysis, chronic progressive external ophthalmoplegia, elevated CK